

HARSH RAJ

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EDUCATION

Northeastern University, Boston, MA	May 2027 (expected)
Master of Science in Computer Science	GPA: 3.8/4.0
Delhi Technological University, Delhi, India	August 2023
Bachelor of Technology in Engineering Physics (Minor in Computer Science)	GPA: 3.4/4.0

TECHNICAL SKILLS

Programming Languages:	Python, SQL, JavaScript, C++, Java
Cloud and DevOps:	AWS, GCP, Docker, Kubernetes, Git, CI/CD
Databases:	PostgreSQL, MySQL, MongoDB, Redis, Vector Databases
Tools and Libraries:	PyTorch, Jax, Hugging Face, Ray, LangChain, Pandas, NumPy, CUDA, Triton

PROFESSIONAL EXPERIENCE

Mahine Learning Research Intern, Scale AI, Manhattan, NYC, NY	May 2026 – Present
• Research on AI Agent Failure Taxonomy.	
Research Engineer Intern, Bespoke Labs AI, Mountain View, CA	Jan 2026 – Apr 2026
• Curated synthetic RL training environments for leading frontier labs (details under NDA).	
• Impact: Work contributed to some of the best proprietary CLI agents in the market.	
Research Assistant, Forschungszentrum Jülich	July 2025 – Present
• Contribute to EU Open Foundational Model projects (MINERVA , OpenEuroLLM , ELLIOT) on JUPITER, EU's largest supercomputer.	
• Lead ablation studies on dataset mixtures and scaling laws on petabyte-scale LLM training data.	
• Impact: Optimized multi-node data processing pipelines using Ray, achieving 5x speedup in deduplication, filtering, and map-reduce operations.	
Co-Founder, Ontocord AI	Jan 2024 – May 2025
• Co-founded open-source initiative focused on creating high-quality permissive datasets for LLMs.	
• Developed data processing pipelines with strategies for mixing, formatting, upsampling, interleaving, and quality validation.	
• Impact: Released MixtureVitae-300BT achieving 10% performance improvement over existing permissive datasets, also matching non-permissive alternatives.	
Applied Scientist, Vijil AI	Aug 2023 – Nov 2024
• Contributed to vijil-evaluation platform for enterprise LLM workflow red-teaming. Integrated 15+ benchmark datasets for vulnerability assessment	
• Developed vijil-fuzzer framework for adversarial text mutation, achieving 85% success rate against GPT-4o at launch	
• Impact: The vijil-evaluate service is used by enterprise clients including DigitalOcean and Groq	

PUBLICATIONS

MixtureVitae: Open Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning Data	<i>preprint</i>
Improving Consistency in Large Language Models through Chain of Guidance 1st Author	TMLR 2025
Defences against Reverse Preference Attacks 3rd Author	AAAI 2025
Transfer of Adversarial Robustness from Pretraining to Downstream 2nd Author	NeurIPS 2023
Measuring Reliability of Large Language Models through Semantic Consistency 1st Author	NeurIPS 2022
<i>Best Paper Award</i>	

PROJECTS

harbor (1500+ GitHub ★)	Language Agent Sandbox Evaluation framework. (Co-author)
terminal-bench (1500+ GitHub ★)	Industry standard Benchmark for CLI Agents. (Co-author)
inspect-ai (1300+ GitHub ★)	LLM Evaluation Suite. (Contributor)
garak (5000+ GitHub ★)	LLM Red-Teaming Suite. (Contributor)

AWARDS AND ACHIEVEMENTS

SWISS AI Grant Recipient – Awarded 200k GPU hours and CHF 61,000 in funding	2025
3rd Place – Vision and Language Navigation R2R Benchmark Competition	2024
ML Safety Challenge – Best Paper Award with a \$5000 prize	2022
Rank 7 – Codeforces Raif ML Round 1 Competition	2021